

EV Charging Network Operational Activities Report

1. Network Reliability & Hardware Fault Detection

Real-Time Activity: Telemetry & Status Code Log Analysis

Industry Scenario: 14% of DC Fast Chargers in Metro North dropped offline/faulted after firmware update.

2. Load Balancing & Utilization Optimization

Real-Time Activity: Spatial-Temporal Capacity & Utilization Mapping

Industry Scenario: Downtown hubs 95% utilized while nearby suburban hubs are 15%.

3. Driver Bottlenecks & Throughput Management

Real-Time Activity: Queue Time & Session Duration Modeling

Industry Scenario: Highway hubs face long queues due to charging beyond 80% SoC.

4. Revenue Leakage & Dynamic Margin Analysis

Real-Time Activity: Unit Economics & Margin Audit

Industry Scenario: Energy sales up 20% but margins down 5% in some regions.

5. Grid Peak-Demand Tariff Mitigation

Real-Time Activity: Smart Grid Power Draw & ToU Cost Analysis

Industry Scenario: Peak demand charges increased operating costs.

6. Predictive Battery & Hardware Health Monitoring

Real-Time Activity: Equipment Degradation & Anomaly Detection

Industry Scenario: Thermal metrics rise before hardware failures.

7. Strategic Capital Allocation for Expansion

Real-Time Activity: Geospatial Heatmap & Expansion Scoring

Industry Scenario: Select best new charging sites using data-driven scoring.

8. Transaction & Payment Gateway Drop-off Audit

Real-Time Activity: Session Initialization & Payment Failure Tracking

Industry Scenario: 8% of charging sessions fail before power delivery.

9. Customer Satisfaction (NPS) Sentiment Mapping

Real-Time Activity: Multi-Channel Review & Telemetry Correlation

Industry Scenario: Poor reviews linked to slow chargers and broken screens.

10. Competitor Pricing & Speed Benchmarking

Real-Time Activity: Market Competitive Intelligence Analysis

Industry Scenario: Competitor offers faster chargers at lower prices.